# MICHAEL (mike) NEUDER

✉ michael.neuder@gmail.com
michaelneuder.github.io
Princeton, New Jersey, USA

## EDUCATION

**Princeton University** — Aug. 2025 - Present
*Doctoral student, Computer Science* — *Princeton, NJ*
↳ *Under the supervision of Prof. Matt Weinberg*
↳ *Supported by an Ethereum Foundation PhD Fellowship (FY25-2276)*

**Harvard University** — Aug. 2020 - May 2021
*Master of Science, Computational Science* — *Cambridge, MA*

**University of Colorado** — Aug. 2015 - May 2020
*Bachelor of Arts, Computer Science* — *Boulder, CO*
*Bachelor of Arts, Mathematics*

## ACADEMIC EXPERIENCE

**EconCS Group, Harvard University** — Aug. 2019 - Aug. 2021
*Research Assistant supervised by Prof. David C. Parkes* — *Cambridge, MA*

**Santa Fe Institute** — Jun. 2018 - Aug. 2018
*Undergraduate Research Fellow: Summer 2018 REU* — *Santa Fe, NM*

**Bradley Lab, University of Colorado** — Apr. 2017 - Aug. 2021
*Research Assistant supervised by Prof. Elizabeth Bradley* — *Boulder, CO*

**Mozer Lab, University of Colorado** — Mar. 2017 - May 2019
*Research Assistant supervised by Prof. Michael Mozer* — *Boulder, CO*

## INDUSTRY EXPERIENCE

**The Ethereum Foundation** — Mar. 2023 - May 2025
*Researcher – Applied Research Group (ARG)* — *New York, NY*

**Google**
*Software Engineer (L3 → L4) – Cloud Storage* — Aug. 2021 - Feb. 2023 (*Cambridge, MA*)
*Software Engineering Intern – Network Infrastructure* — Summer 2020 (*remote*)
*Software Engineering Intern – Flights* — Summer 2019 (*Cambridge, MA*)
*Software Engineering Intern – Mobile Device Management* — Fall 2018 (*Sunnyvale, CA*)

**Lockheed Martin & LASP**[1] — Feb. 2017 - Oct. 2017
*Software Engineering Intern* — *Boulder, CO*

## JOURNAL AND CONFERENCE PUBLICATIONS (REVERSE CHRONOLOGICAL)

- M. Bahrani, **M. Neuder**, S. M. Weinberg, "Adversarial procurement in blockchains," (2026) *ACM Conference on Computer and Communications Security (CCS'26)*. https://arxiv.org/pdf/2605.05559.
- P. Garimidi, **M. Neuder**, T. Roughgarden, "Beyond Winner-Take-All Procurement Auctions," (2026) *Financial Cryptography and Data Security (FC'26)*. https://arxiv.org/pdf/2603.27779.
- M. Bahrani, **M. Neuder**, S. M. Weinberg, "Selfish mining under general stochastic rewards," (2025) *Advances in Financial Technologies (AFT'25)*. https://doi.org/10.4230/LIPIcs.AFT.2025.20. (full version on arxiv: https://arxiv.org/abs/2502.20360.)

[1] Laboratory of Atmospheric and Space Physics: https://lasp.colorado.edu/home/.

- **M. Neuder**, M. Pai, and M. Resnick, "Optimizing Exit Queues for Proof-of-Stake Blockchains: A Mechanism Design Approach," (2024) *Advances in Financial Technologies (AFT'24)*. https://doi.org/10.4230/LIPIcs.AFT.2024.20.
- Z. Fan, F. Marmolejo-Cossio, D. J. Moroz, **M. Neuder**, R. Rao, and D. C. Parkes, "Strategic Liquidity Provision in Uniswap v3," (2023) *Advances in Financial Technologies (AFT'23)*. https://doi.org/10.4230/LIPIcs.AFT.2023.25.
- **M. Neuder**, E. Bradley, E. Dlugokencky, J. W. C. White, J. Garland, "Detection of Local Mixing in Time Series using Permutation Entropy," (2021) *Physical Review E 103*. https://doi.org/10.1103/PhysRevE.103.022217.
- **M. Neuder**, D. J. Moroz, R. Rao, D. C. Parkes, "Defending Against Malicious Reorgs in Tezos Proof-of-Stake," (2020) *ACM Conference on Advances in Financial Technologies (AFT'20)*. https://doi.org/10.1145/3419614.3423265.
- **M. Neuder**, D. J. Moroz, R. Rao, D. C. Parkes, "Selfish Behavior in the Tezos Proof-of-Stake Protocol," (2020) *Cryptoeconomic Systems (CES'20) Conference*. https://arxiv.org/pdf/1912.02954.
- J. Garland, T. Jones, **M. Neuder**, J. W. C. White, E. Bradley, "An information-theoretic approach to extracting climate signals from deep polar ice cores," (2019) *Chaos: An Interdisciplinary Journal of Nonlinear Science* 29:101105. https://doi.org/10.1063/1.5127211.
- J. Garland, T. Jones, **M. Neuder**, V. Morris, J. W. C. White, E. Bradley, "Anomaly Detection in Paleoclimate Records using Information Theory," (2018) *Entropy* 20(12):931. https://doi.org/10.3390/e20120931.

## WORKSHOP PAPERS AND LETTERS

- P. Garimidi, **M. Neuder**, T. Roughgarden, "Tullock Contests in the Wild: Applications in Blockchains," (2025) *ACM SIGecom Exchanges, Vol. 23, No. 1*. https://www.sigecom.org/exchanges/volume_23/1/GARIMIDI.pdf.
- **M. Neuder**, D. J. Moroz, R. Rao, D. C. Parkes, "Low-cost attacks on Ethereum 2.0 by sub-1/3 stakeholders," (2020) *Workshop on Game Theory in Blockchain, Conference on Web and Internet Economics (WINE)*. https://arxiv.org/pdf/2102.02247.

## ACADEMIC TALKS

- **Beyond Winner-Take-All Procurement Auctions**; *FC'26.* slides.
- **Selfish mining under general stochastic rewards**; *AFT'25.* recording. slides.
- **Optimizing Exit Queues for Proof-of-Stake Blockchains**; *AFT'24.* slides.
- **Low-cost attacks on Ethereum 2.0 by sub-1/3 stakeholders**; *GTiB@WINE'20.* recording. slides.
- **Defending Against Malicious Reorgs in Tezos**; *AFT'20.* recording. slides.
- **Selfish Behavior in the Tezos PoS Protocol**; *CES'20.* recording. slides.
- **Animal Tracking using Deep Learning**; *SFI REU'18.* recording.

## AWARDS & FUNDING

**2025-2030** Ethereum Foundation PhD Fellowship (award FY25-2276; $200,000).
**2020** Computer Science Discovery Learning Award (University of Colorado).
**2019** Sieglinde Talbott Haller Scholarship in Mathematics (University of Colorado).
**2019** Honorable Mention: Computing Research Assoc. Outstanding Undergraduate Researcher. link.
**2017** Phi Beta Kappa (University of Colorado). link.
**2015-2020** President Joseph A. Sewall Esteemed Scholar Award (University of Colorado). link.
**2015-2020** Dean's List (University of Colorado). link.

## PROFESSIONAL SERVICE

Columbia CryptoEconomics Workshop (CCE); **2023, 2024, 2025.** *Organizing Committee.* link.

ACM Conference on Economics and Computation (EC); **2026.** *Program Committee.* link.
Designing Defi ($D^2$) Conference; **2026.** *Program Committee.* link.
The Latest in DeFi Research (TLDR) Conference; **2024, 2025.** *Program Committee.* link.
Crypto & Blockchain Economics Research Forum Conference; **2025, 2026.** *Program Committee.* link.
Workshop on Decentralized Finance (DeFi) at Financial Cryptography; **2026.** *Program Committee.* link.
Princeton Blockchain Special Interest Group (SIG); **2025, 2026.** *Co-organizer.*
Symposium On Discrete Algorithms (SODA); **2026.** *External Reviewer.*
ACM Symposium on Principles of Distributed Computing (PODC); **2026.** *External Reviewer.*
Crypto Academic Summer School at Edge City; **2024.** *Organizing Committee.* link.
dYdX MEV Committee; **2023.** *Committee Member.* link.

## BLOCKCHAIN ARTICLES (NON-PEER REVIEWED) (BY TOPIC)

### Maximal Extractable Value (MEV)

Enshrined Proposer-Builder Separation (ePBS)

- ePBS, distilled. link. tweet.
- Decoupling ExecutionPayload validity from Data Availability. link. tweet.
- Consider the ePBS. link. tweet.
- ePBS – the infinite buffet. link. tweet.
- Relays in a post-ePBS world, *with Jon, Hasu, Tomasz, Chris, Toni.* link. tweet.
- Payload-timeliness committee (PTC), *with Francesco D'Amato.* link. tweet.
- Why enshrine Proposer-Builder Separation, *with Justin Drake.* link. tweet.
- Equivocation attacks in mev-boost and ePBS, *with Francesco D'Amato.* link. tweet.

Relays

- Bid cancellations considered harmful, *with Thomas Thiery.* link. tweet.
- Optimistic relays and where to find them, *with Ankit Chiplunkar.* link. tweet.
- An optimistic weekend. link. tweet.
- Towards enshrined PBS – an optimistic roadmap. link.
- Optimistic Relay Proposal, *with Justin Drake.* link.

Censorship resistance

- EIP-7547: Inclusion lists. link.
- The Case for ILECTRA. link. tweet.
- Inclusion Lists PoC Specification. link. tweet.
- Inclusion lists: execution, consensus, & engine spec overview. link. tweet.
- Unconditional inclusion lists, *with Toni Wahrstatter.* link. tweet.
- Resistance is not futile; CR in mev-boost. link. tweet.
- No free lunch – a new inclusion list design, *with Vitalik Buterin.* link. tweet.

Enshrined auction mechanisms

- Mechan-stein (alt. Franken-ism). link. tweet.
- On block-space distribution mechanisms, *with Pranav Garimidi, Tim Roughgarden.* link. tweet.
- Execution Tickets, *with Justin Drake.* link. tweet.
- How I learned to stop worrying and love mev-burn, *with Justin Drake, Toni Wahrstatter.* link. tweet.

Execution layer

- Ethereum PropAMMs: today and tomorrow, *with Maryam Bahrani.* link. tweet.
- Proprietary AMMs and Ethereum, *with Maryam Bahrani.* link. tweet.
- Application-controlled execution: A case study on cancel prioritization, *with Maryam Bahrani.* link. tweet.

### Proof-of-Stake

Ethereum consensus

- Timing Games: Implications and Possible Mitigations, *with Caspar Schwarz-Schilling.* link. tweet.
- Concurrent Block Proposers in Ethereum, *with Max Resnick.* link. tweet.
- Rollup-Centric Roadmap (2024 version), *with Alex Stokes.* link. tweet.
- A set-theoretic view of Ethereum coteries. link. tweet.
- Time, slots, and the ordering of events in Ethereum, *with Georgios Konstantopoulos.* link. tweet.

Procurement / zkEVM for Ethereum

- A native zkEVM scales bandwidth, not just execution — link. tweet.
- Capacity Oracles, *with Maryam Bahrani.* — link. tweet.
- On Ethereum Prover Market Design, *with Maryam Bahrani.* — link. tweet.
- On incentivizing anonymous participation, *with Maryam Bahrani.* — link. tweet.

`ETH` the asset

- Issuance Issues – Initial Issue. — link. tweet.
- Issuance Issues – Subsequent Soliloquy. — link. tweet.
- Issuance Issues – Tertiary Treatise. — link. tweet.
- My (e)thesis: settlement, data availability, execution – in that order. — link. tweet.

Exit/withdrawal queues

- EIP-7922: Dynamic exit queue rate limit, *with Mikhail, Mallesh.* — link.
- ELI5: Ethereum Validator Exits, *with Mallesh Pai.* — link. tweet.
- Adding flexibility to Ethereum's exit queue, *with Mikhail Kalinin, Mallesh Pai.* — link. tweet.

EIP-7251; `MAX_EFFECTIVE_BALANCE`

- EIP-7251: Increase the MAX_EFFECTIVE_BALANCE. — link.
- Validator consolidation in EIP-7251, *with Francesco D'Amato, Mikhail Kalinin.* — link. tweet.
- Slashing penalty analysis; EIP-7251, *with Barnabe Monot.* — link. tweet.
- FAQ on EIP-7251, *with Francesco D'Amato, Mikhail Kalinin, dAppLion.* — link. tweet.
- A modest proposal, *with Francesco D'Amato, Aditya Asgaonkar, Justin Drake.* — link. tweet.

Liquid staking / restaking

- The Risks of LRTs, *with Tarun Chitra.* — link. tweet.
- Magnitude and direction of Lido attack vectors. — link. tweet.
- Musings on "two-tiered" staking, a native Liquid Staking Token design. — link. tweet.

Blobs

- Blob gossip and validation before and after PeerDAS. — link. tweet.
- On the future of the blob mempool, *with Julian Ma.* — link. tweet.

**Random**

- Overnight reverse repurchases and the Fed's balance sheet. — link. tweet.

## BLOCKCHAIN PRESENTATIONS, PODCASTS, PANELS (REVERSE CHRONOLOGICAL)

### Presentations

- How to get untrusted people to work for you – *Princeton DeCenter Spring Conference. April 2026.* slides.
- Blockchain governance in practice – *EGR 373 (Princeton) Guest talk. March 2026.* slides.
- Selfish Mining under General Stochastic Rewards – *IC3 Winter Retreat. January 2026.* slides.
- Adversarial procurement – *Columbia CryptoEconomics (CCE). December 2025.* recording. slides.
- First Principles of Blockchains – *World Bank Session. November 2025.* slides.

EthCC[8] (July 2025) – Cannes, France

- ↳ Blobs and the future of the blob mempool – *EthCC main event.* recording. slides.
- ↳ Blob futures and the blob mempool – *Blockspace Futures Day.* recording. slides.

- Revisiting Ethereum's Exit Queue – *Beam Chain Call #4. April 2025.* recording. slides.
- Validator incentives – *Oxford-Harvard Conference on DeFi. January 2025.* slides.

DEVCON 7 (November 2024) – Bangkok, Thailand

- ↳ `ETH` is permissionless money – *DEVCON Main stage.* recording. slides.
- ↳ The ticker is `ETH` – *Bankless Summit.* recording. slides.

- Neutrality in Ethereum – *Harvard Club of New York City. October 2024.* slides.
- Separating MEV and Staking Rewards – *CBER Webinar. September 2024.* recording. slides.
- More slides about Ethereum Blockspace – *Scroll Protocol Symposium. August 2024.* recording. slides.
- More slides about PBS – *MEV Workshop, Stanford Blockchain Conf. August 2024.* recording. slides.
- Postmodern Staking – $9^{th}$ *Annual IC3 Blockchain Camp. June 2024.* slides.
- Is restaking the new staking? – *CBER Forum Annual Conference. May 2024.* recording. slides.
- State of the Ethereum Union ($1/N$ perspective) – *ETHBoston. April 2024.* slides.

ETHDenver (March 2024) – Denver, USA

- ↳ Execution Tickets – *Beyond the Block (Titan Builder & Hashkey Capital).* recording. slides.
- ↳ The Risks of LRTs – *Research Day (SevenX Ventures).* recording. slides.

- No Free Lunch – *Columbia CryptoEconomics Working Session. December 2023.* recording. slides.
- Enshrining PBS – *Center for Digital Finance & Technologies. December 2023.* slides.

DevConnect (November 2023) – Istanbul, Turkey

- ↳ Increase the MAX_EFFECTIVE_BALANCE – *EthStaker Staking Gathering.* recording. slides.
- ↳ A set theoretic view of Ethereum coteries – *LidoConnect.* recording. slides.

- Why it's hard to enshrine PBS – *Archetype MEV Lunch. November 2023.* recording. slides.

EthCC[6] (July 2023) – Paris, France

- ↳ Towards Enshrined Proposer-Builder Separation – *EthCC main event.* recording. slides.
- ↳ Ethereum PBS R&D Roadmap – *Modular Summit.* recording. slides.
- ↳ Increase the MAX_EFFECTIVE_BALANCE – *Kiln Rendez-vous.* recording. slides.

- Reorgs in PoS – *MEV Roast; Reorg Edition. August 2021.* recording. slides.

### Podcasts

- Understanding ePBS through the Lens of Game Theory – *Ready for Merge. October 2026.* recording.
- Ethereum - ePBS, PropAMMs, ETH – *Credible Commitments. October 2026.* recording.
- Pectra Explained – *Coinbase Institutional Webinar. May 2025.* recording.
- ETH is Permissionless Money – *Cryptoria. December 2024.* English. Chinese.
- Ethereum's North Star – *The Gwart Show. October 2024.* recording.
- The Future of Ethereum: Is This The Right Track? – *Bankless. October 2024.* recording.
- The Ethereum Roadmap is NOT Off Track! – *Bankless. September 2024.* recording.
- ETH Insights: Discussing MaxEB – *Coinbase Webinar. April 2024.* recording. article.
- PeepAnEIP: EIP-7547 – *Ethereum Cat Herders. April 2024.* recording.
- Endgame 2.0: A Guide to Vitalik's Ethereum Roadmap – *Bankless. February 2024.* recording.
- We're Pretty Sure Mike and Max Can Fix MEV – *The Gwart Show. February 2024.* recording.
- PeepAnEIP: EIP-7251 – *Ethereum Cat Herders. February 2024.* recording.

- Eigenlayer In 2024 (co-host) – *Bankless. December 2023.* recording.
- An Incomplete Guide to PBS – *Uncommon Core 2. September 2023.* recording.

**Panels**

- Decentralization and pragmatism – *Princeton DeCenter Spring Conference. April 2026.*
- Restaking Panel – *Staking Rewards: Staking Summit. November 2024.* recording.
- Decentralization Panel – *Espresso: Beyond the Baselayer. March 2024.* recording.
- Decentralize or Bust – *bloXroute & ETHStaker. March 2024.* recording.
- Alignment Panel – *EigenLayer Restaking Summit. November 2023.* recording.
- PBS & Beyond – *MEVDay Paris. July 2023.* recording.

## OPEN SOURCE CONTRIBUTIONS

EIP-7547: Inclusion Lists (`python`) (repo)

- EIP (Ethereum Improvement Proposal). link.
- Spec Overview (Consensus, Execution, and Engine API specifications). link.
- Proof-of-Concept Specification. link.
- Compilation of Related Work. link.

EIP-7251: Increase the `MAX_EFFECTIVE_BALANCE` (`python`) (repo)

- EIP (Ethereum Improvement Proposal). link.
- Minimal Spec Change (Consensus specifications). link.
- Compilation of Related Work. link.

MEV-boost Relay: Optimistic Processing (`golang`) (repo)

- Main pull request & design documentation. link.
- Header-only parsing optimization. link.
- Pull request list. link.
- Proposal. link.
- Builder Onboarding. link.

Erigon (Ethereum Execution Layer Client) (`golang`) (repo)

- Refactored req/resp domain network encoding. link.
- Consensus spec implementation. link.
- Pull request list. link.

Prysm (Ethereum Consensus Layer Client) (`golang`) (repo)

- Keymanager code-health refactor. link.
- Pull request list. link.

STRUMPACK (High-performance matrix math package) (`C++`) (repo)

- Mixed-precision (`float64` & `float32`) iterative refinement. link.
- Pull request list. link.
- Write-up. link.

OptRBC (Optimal solutions in 2D Rayleigh-Benard Convection) (`FORTAN`) (repo)

- Multiprocessor implementation using openmp. link.
- Write-up. link.

Image quality convolutional neural networks. (`python`) (repo)

- Write-up. link.
- Example network. link.